

SAI CHARAN MERUGU

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Experience

Arizona State University <i>Artificial Intelligence Researcher</i>	June 2025 – Present <i>Arizona , USA</i>
- Developed an interactive VR-based cognitive therapy system using AI-driven avatars and transformer-based dialogue models. - Built modular FastAPI services and vector search with FAISS/Qdrant for real-time LLM inference and session handling. - Integrated backend with Unreal Engine, allocating vGPUs to ensure scalable, low-latency response for multi-user sessions.. - Deployed and tested research infrastructure in an Azure environment to support backend inference systems - Maintained persistent session tracking and adaptive feedback loops via MongoDB and behavioral logging.	
Zof AI <i>AI Software Developer Intern</i>	June 2025 – Present <i>San Francisco, USA</i>
- Developed intelligent, self-orchestrated LLM agents for automated regression testing using schema-driven APIs. - Built dual-mode agent communication layers (REST & MCP) using FastAPI and Pydantic with scalable containerized deployment. - Maintained high availability through Docker containers and PM2-based process management on Azure infrastructure. - Used GitHub Actions and Agile tracking tools for CI/CD and iterative model updates.	
University of North Texas <i>AI Research Assistant</i>	March 2024 – May 2025 <i>Denton, Texas USA</i>
- Developed a Retrieval-Augmented Generation (RAG) system for personalized virtual therapy using FAISS, Qdrant, and BERT. - Designed real-time interaction APIs between transformer models and VR therapy applications using FastAPI and vector search. - Integrated Azure-backed compute systems with Unreal Engine and Pixel Streaming, reducing latency by 40%. - Enabled adaptive model responses with personalized feedback loops, managing load through GPU balancing per session. - Supported deployment and monitoring in Azure cloud environments for research use.	
Infosys Ltd <i>Digital Specialist Engineer</i>	August 2022 – August 2023 <i>Hyderabad, India</i>
- Developed and deployed real-time ML pipelines using PySpark and Scala on Azure Databricks for logistics forecasting. - Built structured data pipelines using Azure Data Factory, PostgreSQL, and MongoDB for production-grade analytics. - Integrated Snowflake for batch model scoring and reporting with SQL-based transformations and optimization. - Automated Python-based model training and microservices, reducing inference latency and boosting reliability. - Containerized services with Docker, and deployed using Jenkins and GitHub Actions under Agile delivery cycles.	
Darwinbox <i>Software Engineer Intern</i>	March 2022 – August 2022 <i>Hyderabad, India</i>
- Built a type-safe frontend using React and TypeScript, improving UI reliability and reducing runtime errors across releases. - Developed backend services with Node.js and Express, implementing authentication and session management using MongoDB, aligned with MEAN architecture principles. - Dockerized the full-stack environment, enabling seamless local development and consistent deployment across environments. - Set up CI/CD pipelines using GitHub Actions, ensuring fast, reliable deployments and streamlined testing workflows.	
Exposys Data Labs <i>Data Engineer Intern</i>	Sep 2021 – Feb 2022 <i>Hyderabad, India</i>
- Built preprocessing pipelines for structured datasets using Python (Pandas, NumPy), enabling smooth data preparation for modeling - Performed exploratory data analysis (EDA) and created visualizations using Matplotlib and Seaborn to uncover trends and patterns	

- Trained and evaluated ML models (classification, regression) using scikit-learn, with performance measured via F1 score and AUC.
- Supported experimental deep learning research and collaborated in Agile sprint cycles with the engineering team.

Orbit Shift

Jan 2020 – July 2020

Data Analyst Intern

Hyderabad, India

- Analyzed structured borrower data and built predictive models (Random Forest, AdaBoost, SVM) for loan approvals.
- Engineered features like DTI ratio and job tenure bins to improve interpretability and model performance.
- Evaluated model outcomes using AUC, F1-score, and accuracy to derive data-driven insights.
- Used Python, Pandas, and scikit-learn for EDA, model tuning, and visualization.

Project Experience

ADK SQLite Agent with FastAPI, Gemini Integration | *Python, ADK, SQLite, Gemini, Pydantic* **June 2025, Github**

- Designed an ADK agent powered by Gemini LLMs and FastAPI that queries and modifies a local SQLite database via a self-hosted MCP server using Pydantic tools and schema-driven automation.

Snowflake Food Delivery analysis | *Snowflake, SQL, Streamlit*

May 2025, Github

- Built real-time order analytics pipeline using Medallion architecture and Snowflake delta ingestion.
- Created stream-based dashboards for Swiggy performance metrics using SQL queries and Streamlit.

Research & Publications

Next Generation Neurophysiological Assessments | *HPCC, AWS, Unreal, Pixel Streaming* **2024, Poster CYPsy27**

- Developed a scalable system using GPU-powered servers with Unreal Pixel Streaming and Unity Render Streaming to deliver high-fidelity, ecologically valid virtual environments for neurophysiological assessments via standard web browsers; enabled low-latency, real-time streaming for cognitive state evaluation without requiring high-end client hardware.

Apple Leaf Diseases Classification Using CNN with Transfer Learning

2022, IJARASET

- Developed a Flask-based web application for automated diagnosis of apple foliar diseases using custom CNN and transfer learning models (InceptionV3, VGG16/19), enabling real-time image uploads and achieving up to 93% classification accuracy.

Technical Skills

Languages: Python, SQL, Scala, TypeScript, JavaScript, Java, C++, Go, Bash

Libraries/Frameworks: PySpark, FastAPI, Pydantic, React, Node.js, Transformers, TensorFlow, PyTorch

Tools/Platforms: Azure Databricks, Azure Data Factory, Snowflake, Docker, GitHub Actions, Jenkins, JIRA

Databases: PostgreSQL, Snowflake, MongoDB, MySQL, MongoDB

Concepts: :Data Pipelines, Feature Engineering, Regression Models, ETL, CI/CD, MLOps, LLMs, RAG, DevOps

Cloud: : Azure (Databricks, Functions, Storage, Monitoring)

Education

University Of North Texas

Aug 2023 – May 2025

Masters in Computer Science, GPA: 3.727

Denton, Texas, USA

Sreenidhi Institute of Science & Technology

July 2018 – June 2022

Bachelors Of Technology in Computer Science, GPA: 7.67/10.0

Hyderabad, India

Certifications

AWS Cloud Practitioner, Google Data Analytics Specialization, Databricks - Generative AI Fundamentals